

Hansadee Rathnayaka

COMPUTER ENGINEERING UNDERGRADUATE, UNIVERSITY OF RUHUNA

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Summary

As a Computer Engineering undergraduate at the University of Ruhuna, I specialize in the convergence of full-stack development, DevOps practices, cloud technologies, and AI-enabled systems. Passionate about crafting scalable, intelligent software that seamlessly integrates frontend innovation with backend efficiency and real-time capabilities, I blend strong programming foundations with leadership in tech communities to create impactful solutions that tackle real-world problems through creative and reliable engineering.

Education

- B.Sc. Eng. Hons. in Computer Engineering** May 2023 – Present
University of Ruhuna, Sri Lanka.
 - CGPA: 3.46** | Relevant Coursework: Software Engineering, DevOps, Cloud Computing, Database Systems, AI, ML
- GCE Advanced Level (Mathematics, Physics, Chemistry)** Grad: 2021
Mahinda Rajapaksha College, Matara, Sri Lanka.
 - 3As** | z-score: 1.917
- Other Courses**
 - Web Application Development**
 - Web Design for Beginners (on Open UOM) [Certificate](#) – March 2026
 - Retrieval Augmented Generation(RAG)**
 - Retrieval Augmented Generation (RAG) (on Coursera) [Certificate](#) – March 2026
 - RAG Course for Beginners (on Simplilearn) [Certificate](#) – March 2026

Projects

- Smart Seat Booking Web for Film Halls** | ReactJS, Node.js, Express, MySQL, RESTful API, Java 2025
 - Developed a full-stack web application for real-time film hall seat reservations with secure user authentication.
 - Engineered a robust backend using Node.js/ Express to manage complete CRUD functionality for users, bookings, and seating data.
 - Designed and optimized a MySQL database schema to handle relational data for complex booking transactions.
 - Integrated frontend and backend systems through high-performance RESTful JSON APIs for a seamless user experience.
- Task Manager System** | React, Node.js, Express, MongoDB, Docker, Jenkins, Terraform, Ansible, AWS 2025
 - Developed a production-ready task management web application with secure user authentication.
 - Built a RESTful API with JWT authentication and containerized the application using Docker.
 - Provisioned AWS EC2 infrastructure using Terraform and automated deployment with a Jenkins CI/CD pipeline.
 - Integrated GitHub, Docker Hub, and Ansible for continuous integration and delivery.
- Famous Book Character Chat Application** | FastAPI, LangChain, ChromaDB, SQLite, React, Vite, NLP, RAG 2025
 - Developed an AI-powered web application that allows users to upload book PDFs and interact with characters through contextual chat.
 - Implemented semantic text processing and character profiling using RAG with ChromaDB to produce responses grounded in source content.
 - Built backend services with FastAPI and LangChain, with SQLite for persistence and caching for improved performance.
 - Designed a React + Vite frontend with persistent chat history for enhanced user experience.
- Jungle Odyssey - Online Camping & Adventure Booking System** | React, Node.js, Express, MySQL, Axios, React Router 2026
 - Developed a full-stack camping accommodation booking system with responsive multi-page navigation and interactive user experience.
 - Implemented comprehensive booking functionality, including secure form validation and database integration for trip management.
 - Utilized React Hooks and a component-based architecture to manage state and create a modular, scalable frontend.

- Integrated Axios for seamless API communication between the React frontend and Node.js/Express backend.
- Designed and managed a MySQL database to handle relational data for trip organizers, bookings, and user profiles.
- **Uber Fare Prediction System** 🚗 | Python, Scikit-learn, Pandas, NumPy, Google Colab, Regression 2026
 - Developed a machine learning system to predict ride-sharing fares by processing trip variables such as pickup/dropoff coordinates, passenger count, and time-based features.
 - Implemented and compared multiple supervised learning models, specifically Linear Regression and Random Forest Regressor, to optimize prediction accuracy.
 - Engineered spatial features using the Haversine formula to calculate trip distances and performed extensive data cleaning to remove outliers and improve model robustness.
 - Evaluated model performance using R^2 and RMSE metrics, achieving high transparency in pricing estimation through comparative analysis.

Volunteer and Leadership Experience

- **REXTRO 2025 - Silver Jubilee Exhibition of Faculty of Engineering** *University of Ruhuna*
Team Lead of Finance and Fundraising Committee 2025
- **IEEE Student Branch - University of Ruhuna** 2023 – 2025
Treasurer of IEEE Computer Society (Executive Committee term 23/25)
 - Vice Chair of the event “Haxtreme 3.0”
 - PR lead of “Red Cypher 2.0” and “Haxtreme 2.0”
- **AIESEC - University of Ruhuna** 2023 – 2024
 - IGV B2B Team Lead Term 24.25
 - IGV B2B Team member Term 23.24

Skills Summary

Programming	Java, C#, JavaScript, C, Python
Frameworks and Lib	React, Express.js, FastAPI, LangChain
Databases	MySQL, MongoDB, SQL Queries
Tools	Git and GitHub, VS Code, Google Colab,
Cloud and DevOps	AWS, Docker, Terraform, Jenkins, Ansible, CI/CD

References

Dr. Kushan Sudheera | PhD (NTU, Singapore), B.Sc.Eng.(Hons)(RUH, SL)
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